EECE 5554 — Lab 4

Dead Reckoning with IMU, GNSS, and LiDAR

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https://github.com/ZhiQ1/EECE5554

Data limitation: RTK Fixed was unavailable in the recorded datasets.
GNSS-referenced plots should not be interpreted as RTK-accuracy validation.

Figure 1 — Walking magnetometer calibration

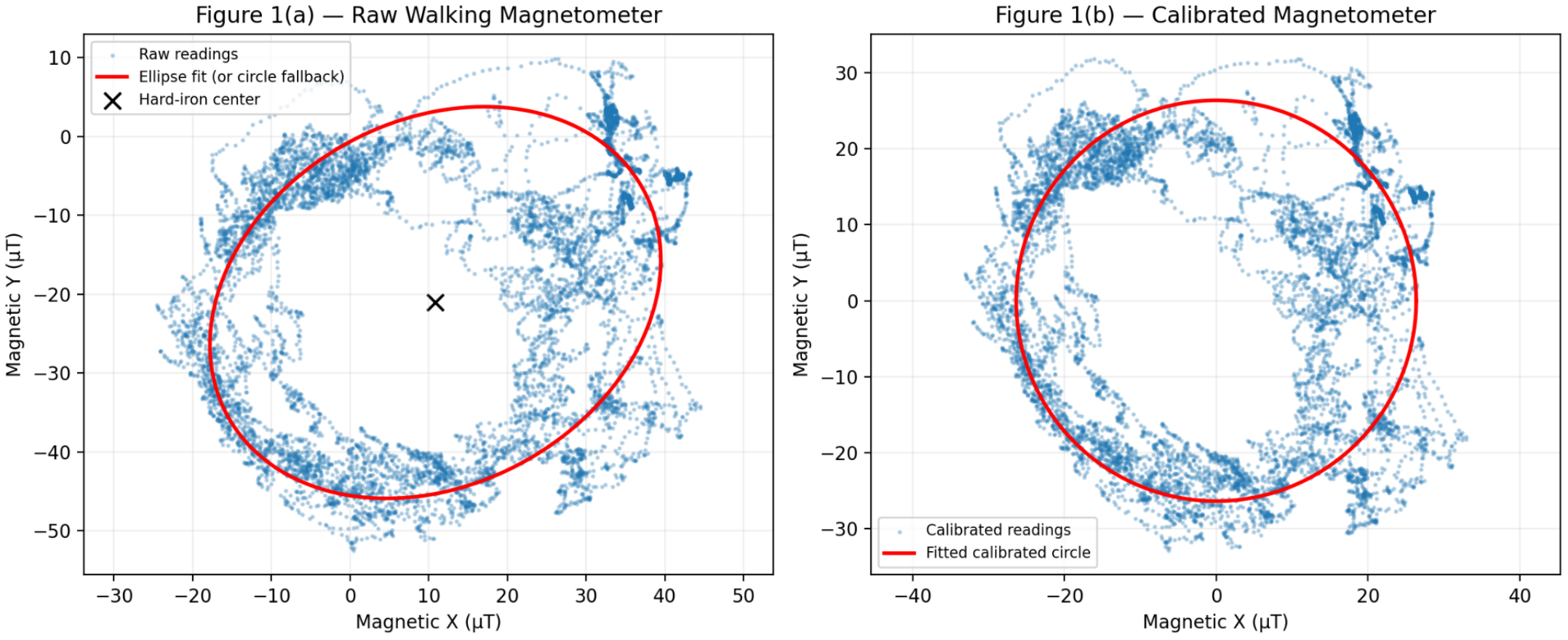


Figure 2 — Walking square heading estimates



Figure 3 — Walking dead reckoning

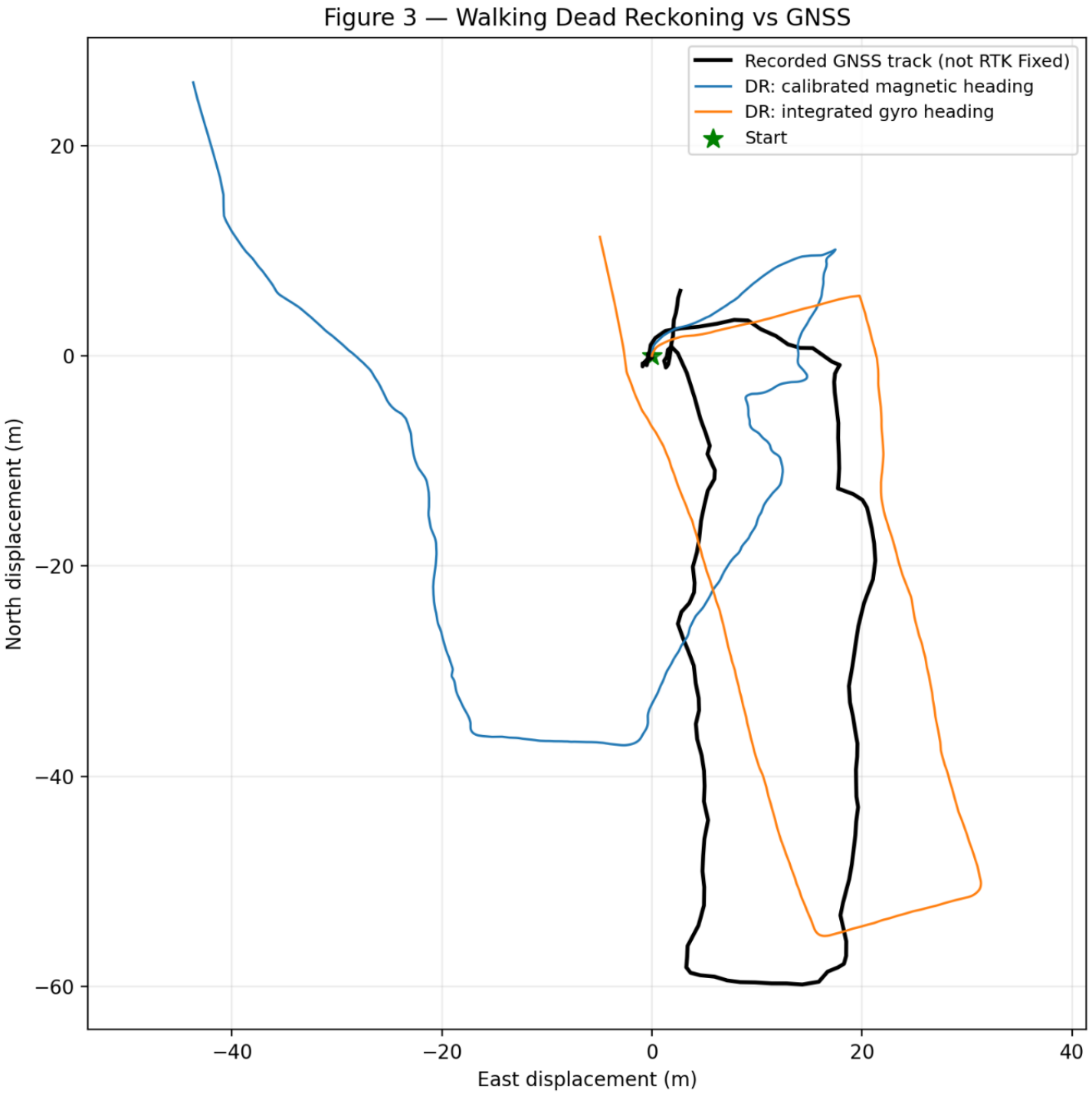


Figure 4 — Car magnetometer calibration

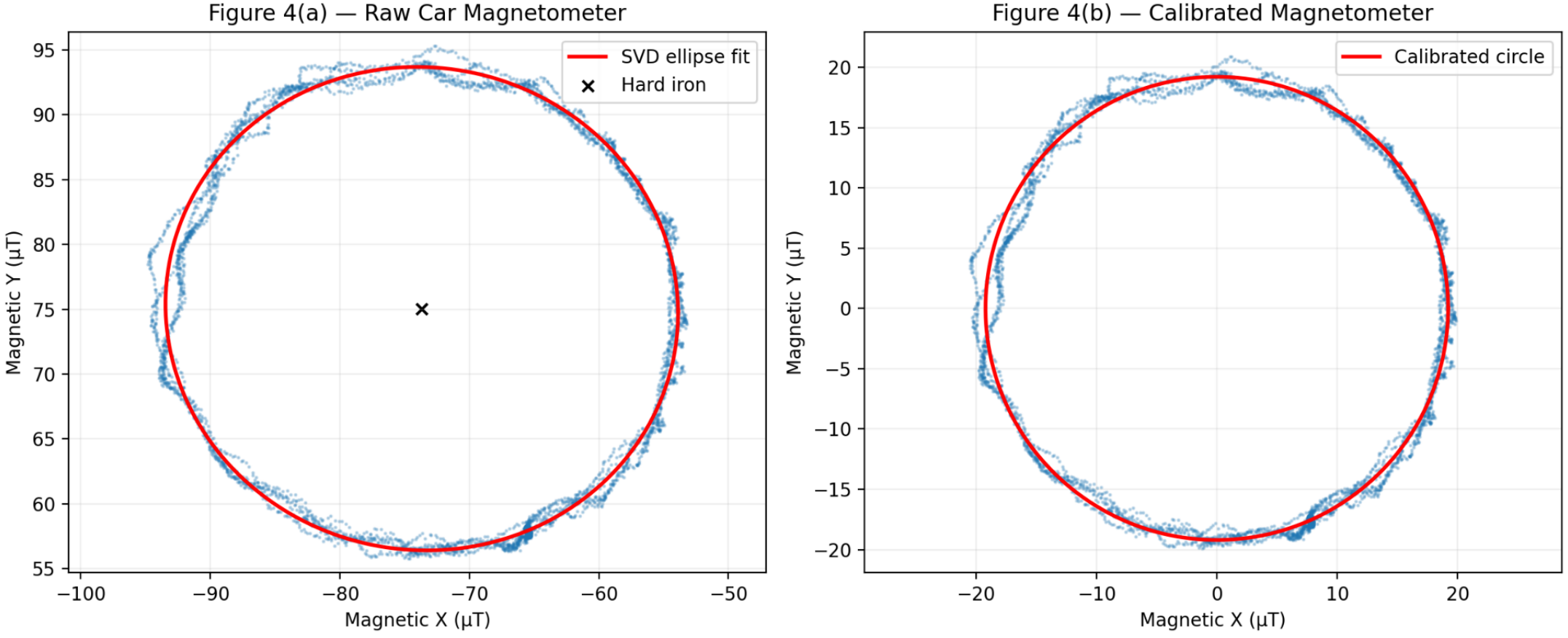


Figure 5 — Car magnetic heading

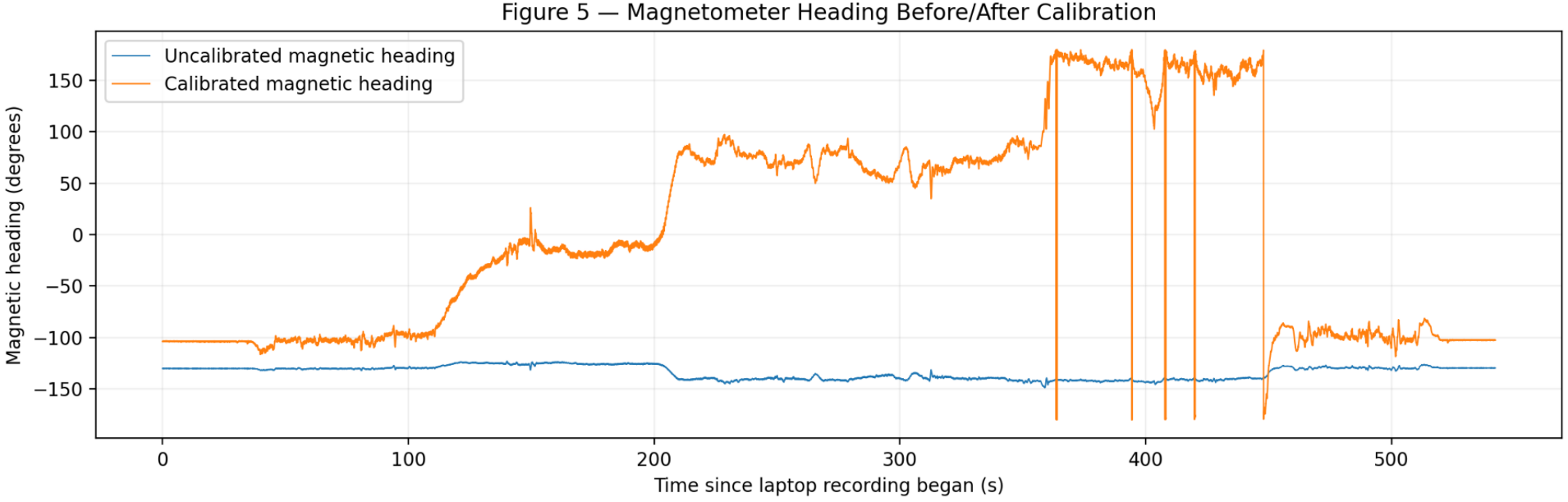


Figure 6 — Gyro vs magnetometer heading

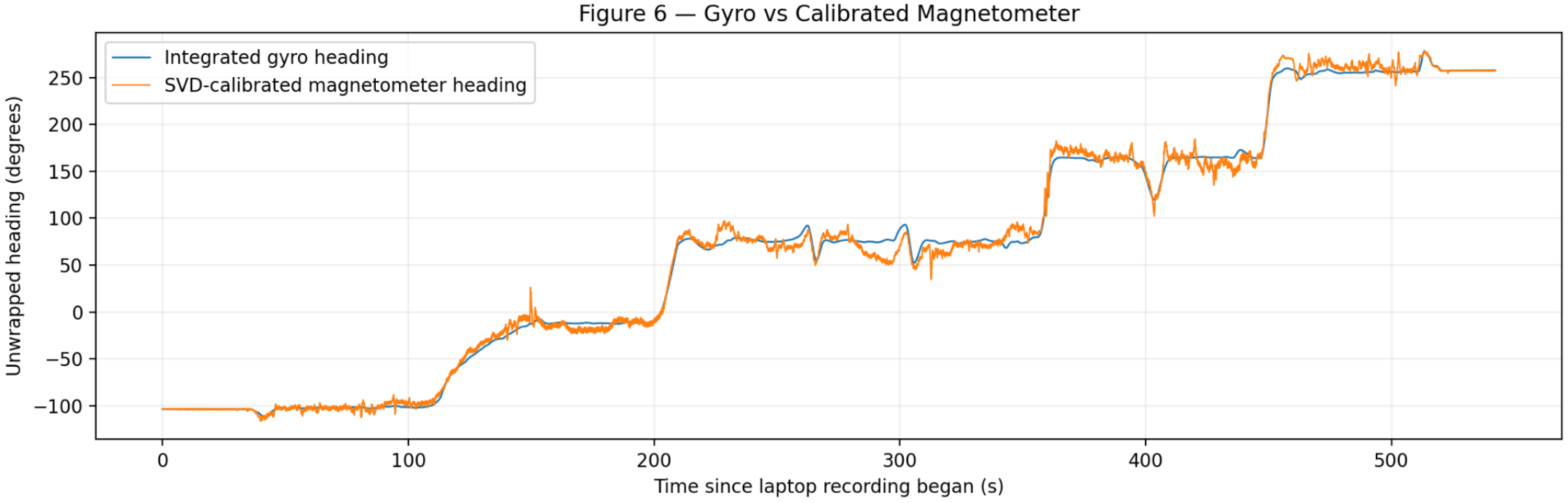


Figure 7 — Complementary filter

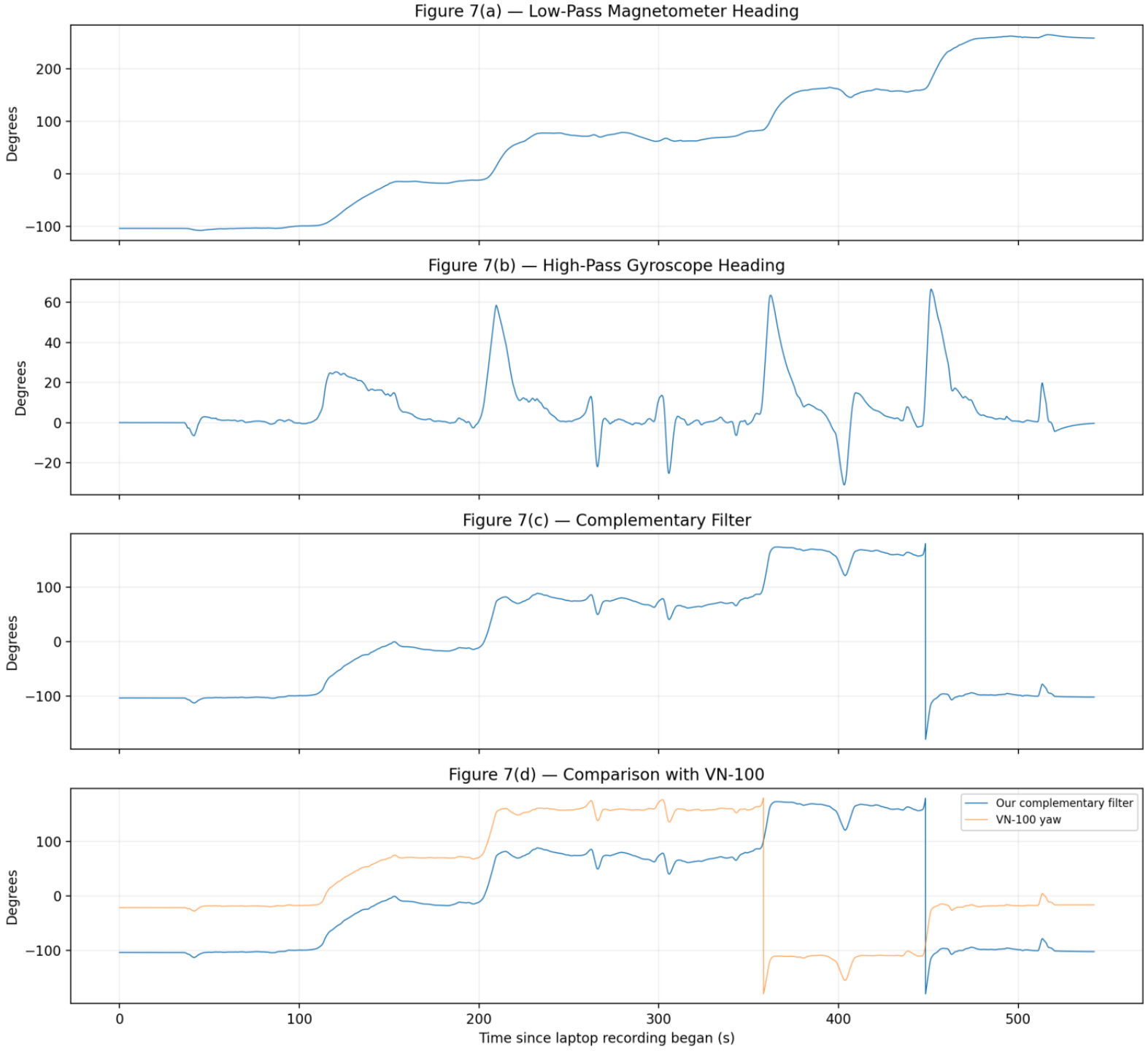
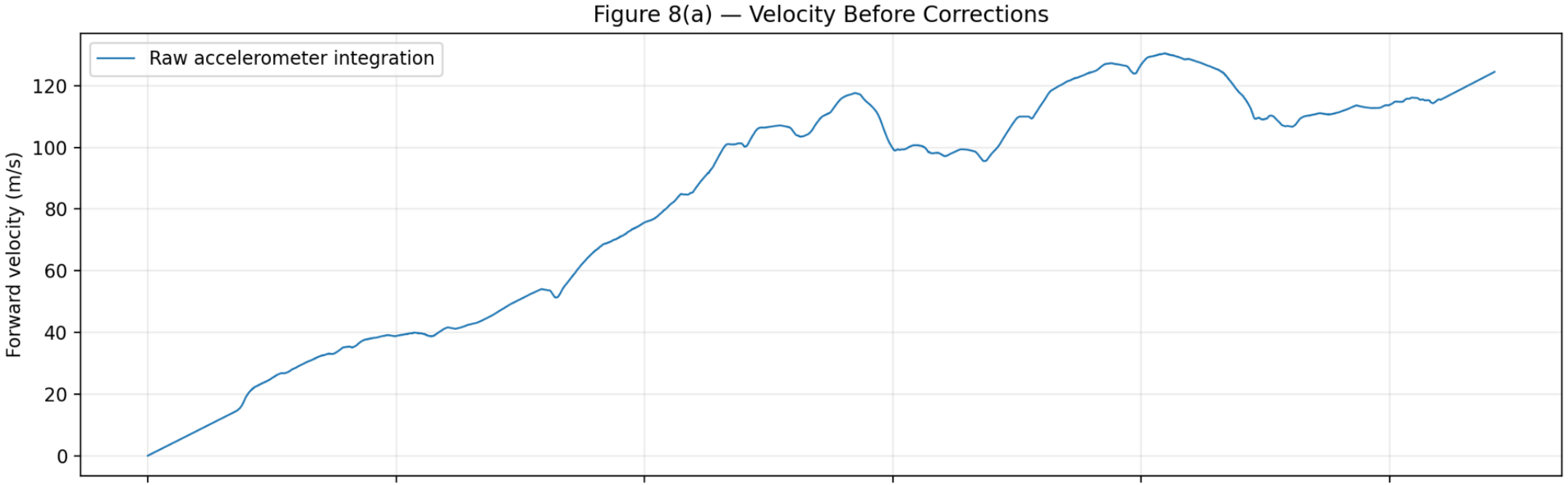


Figure 8 — Forward velocity



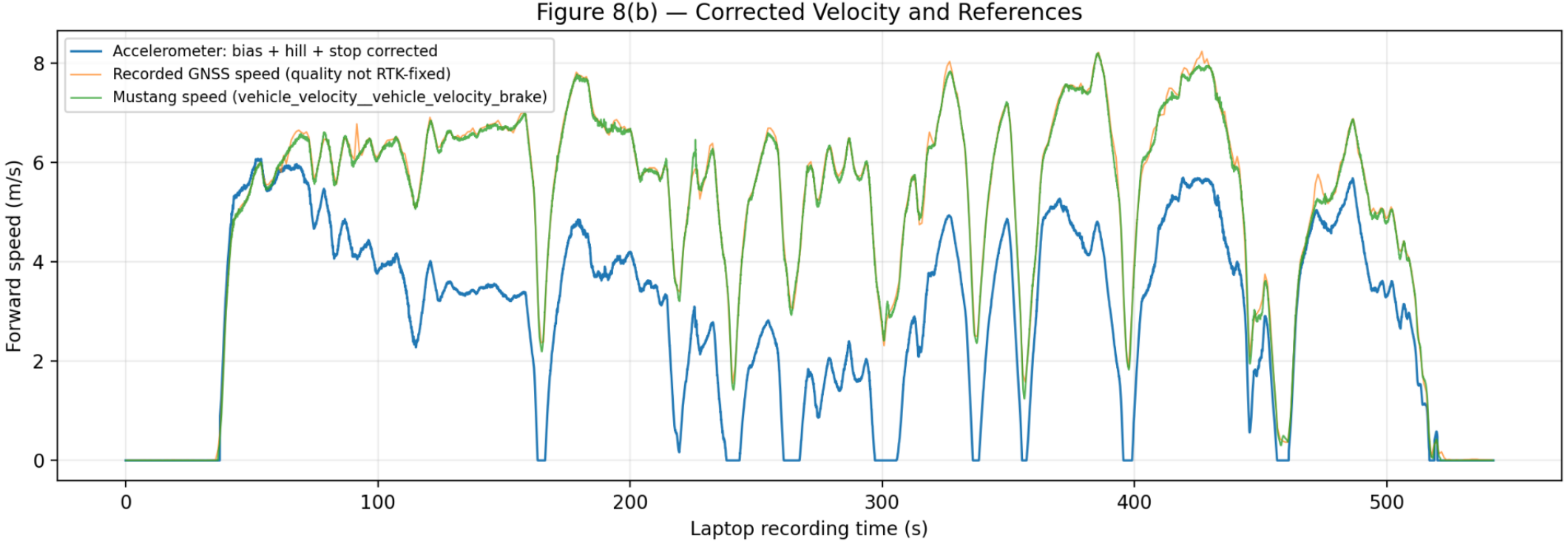


Figure 9 — Vehicle motion model

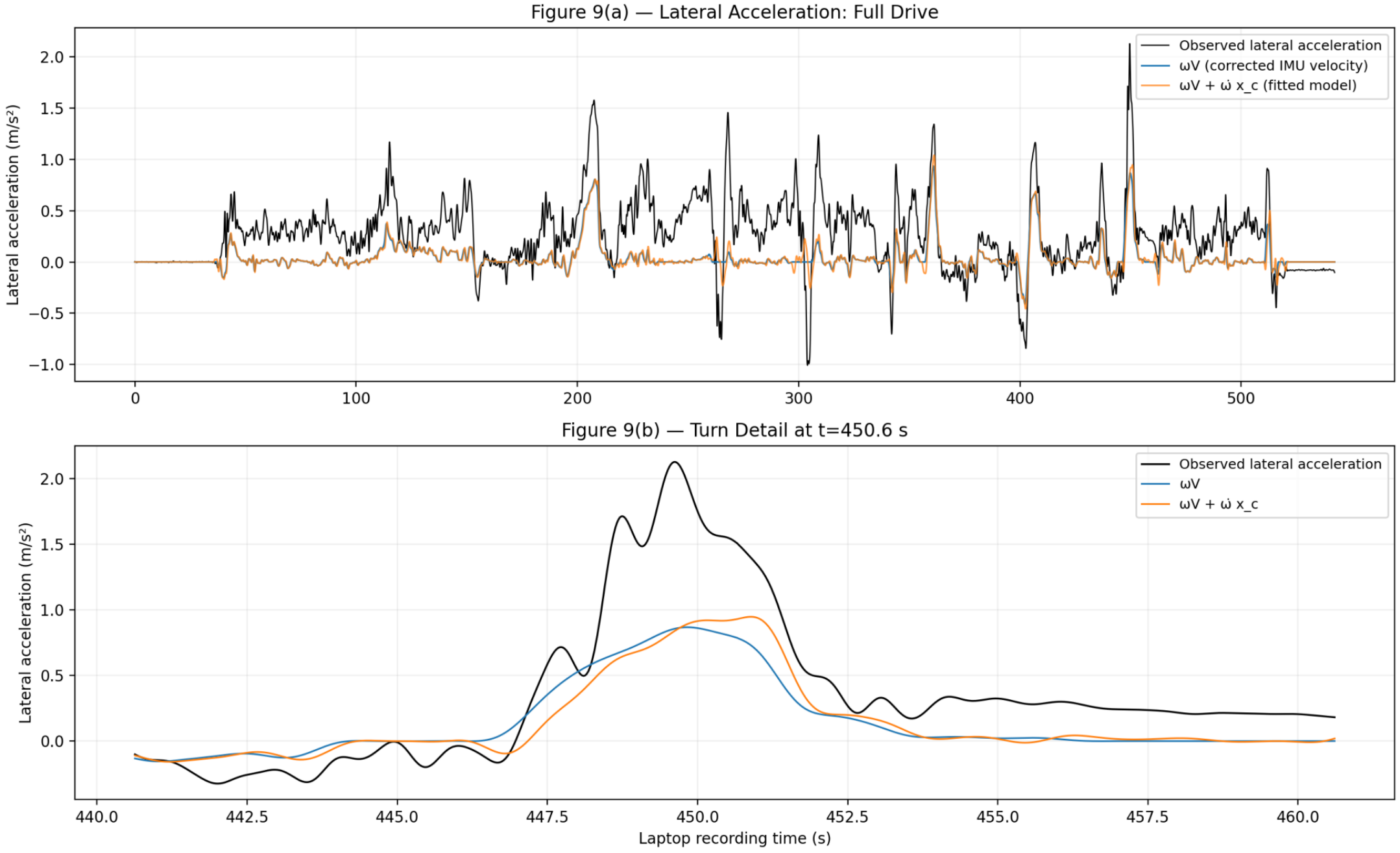
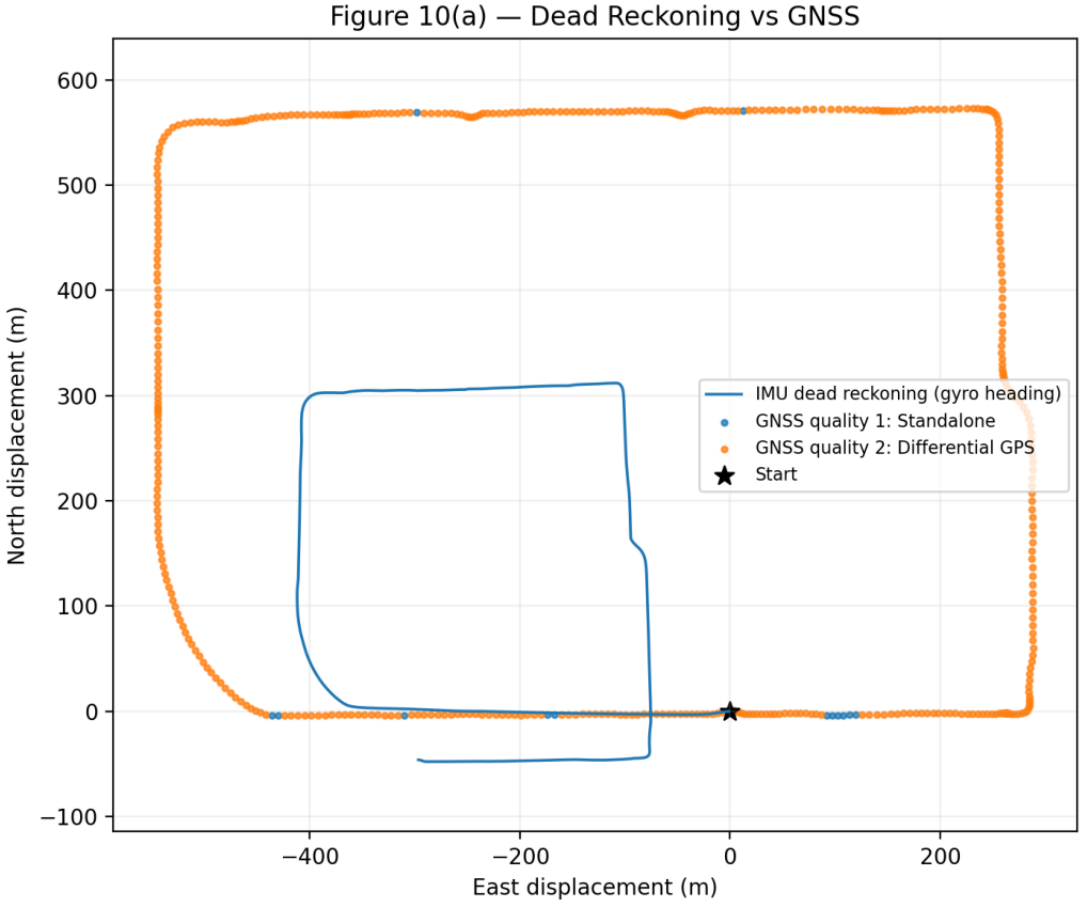


Figure 10 — Dead reckoning vs GNSS



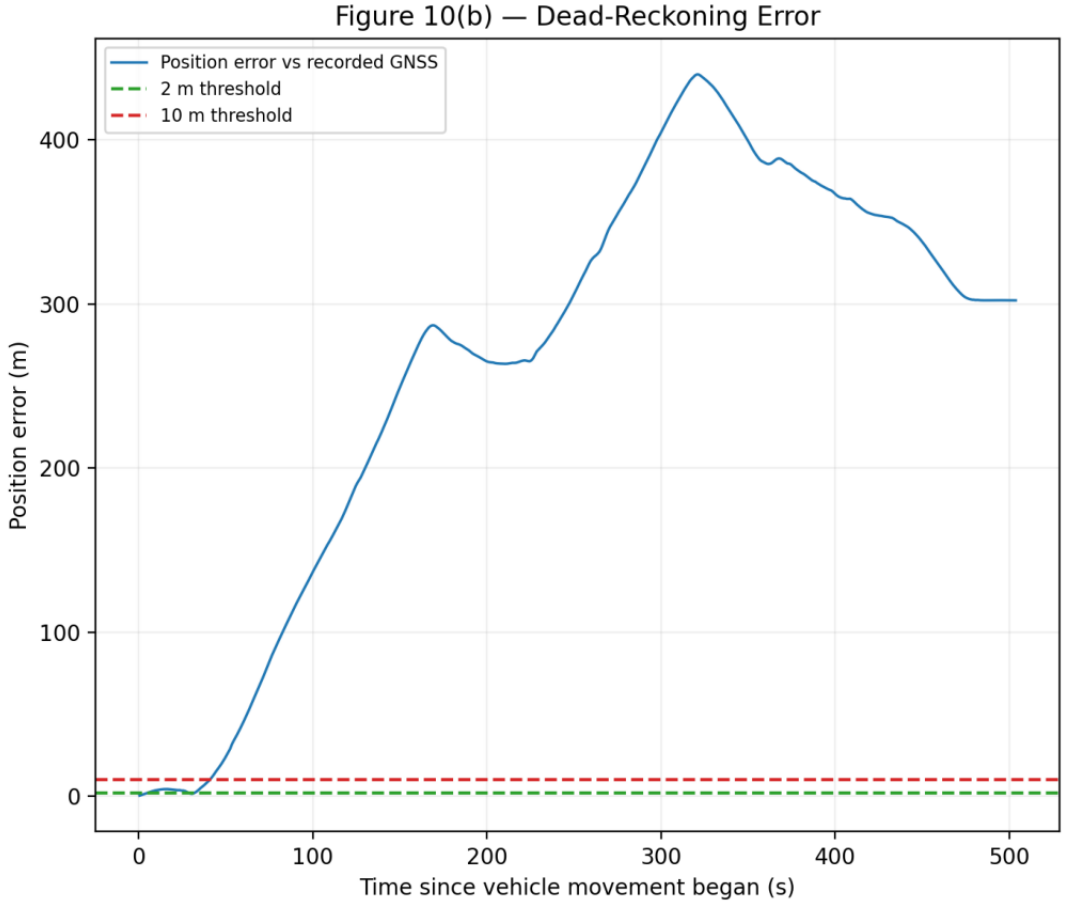


Figure 11 — Two-computer clock alignment

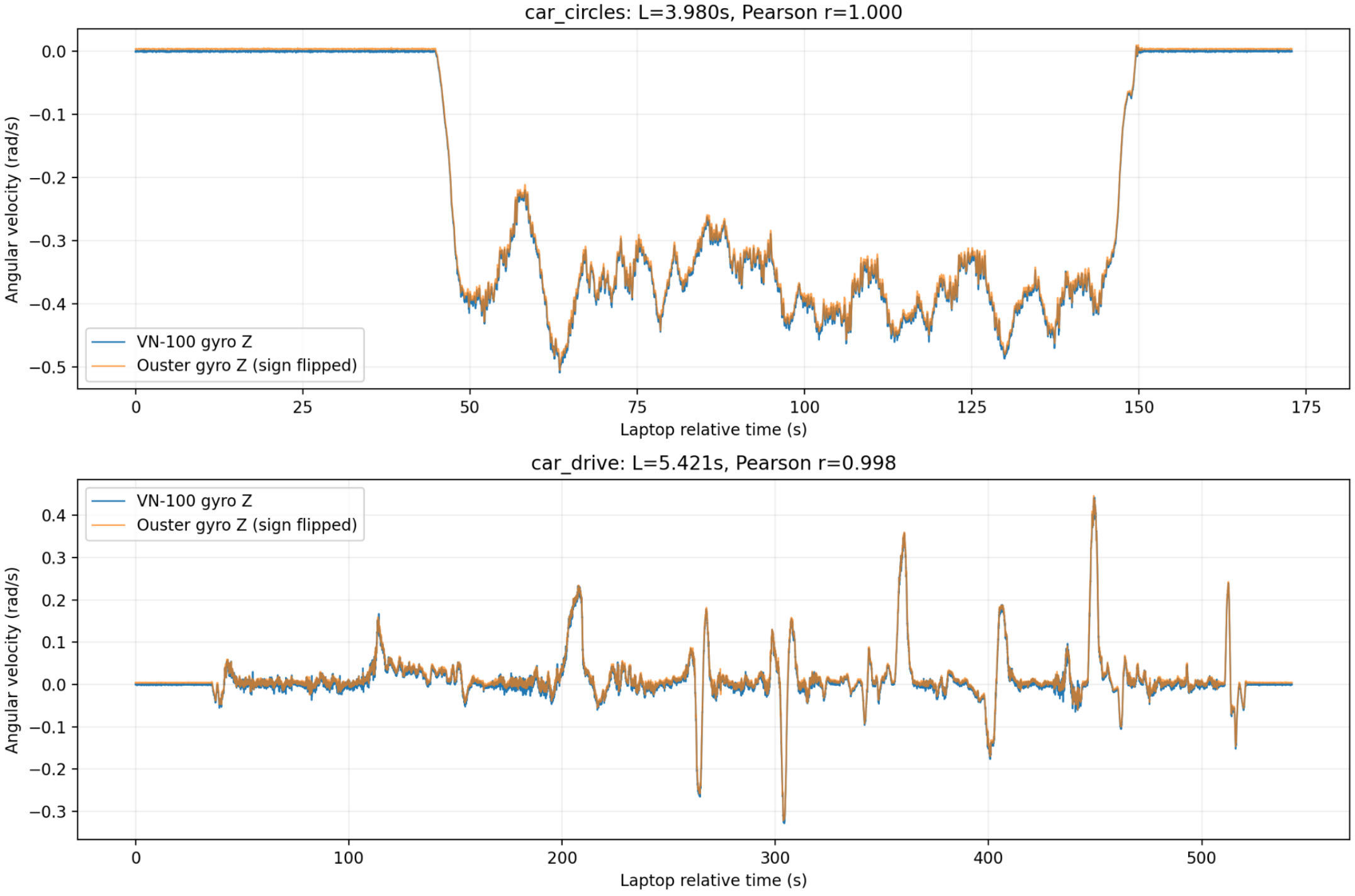
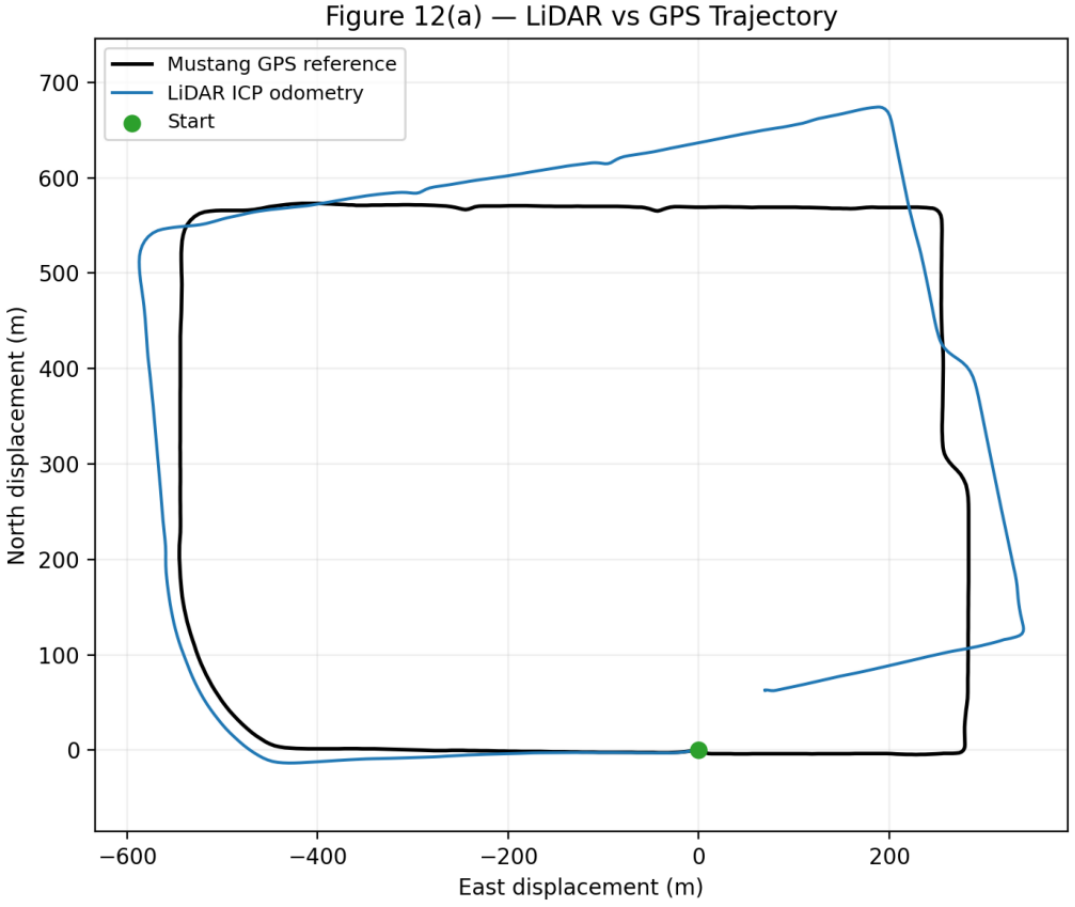


Figure 12 — LiDAR ICP vs GPS



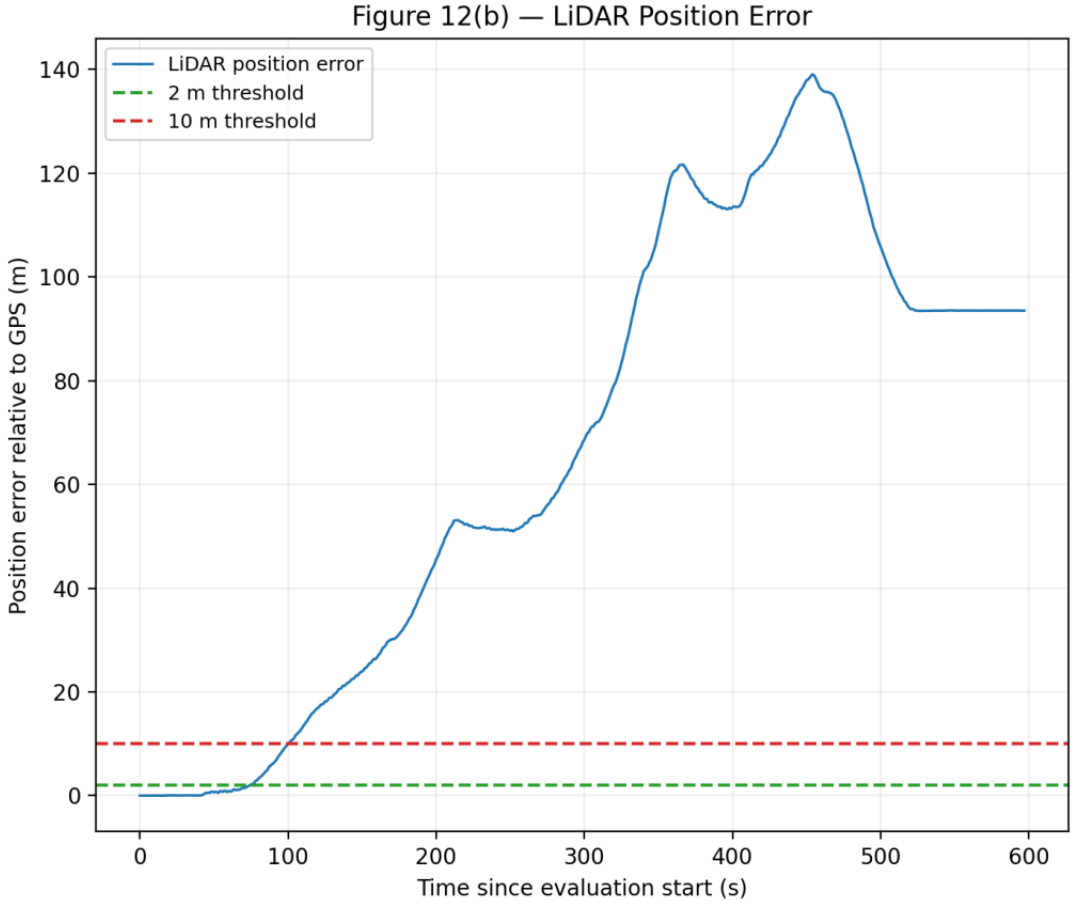


Figure 13 — Simulated 90-second GNSS outage

